

ORIGINAL ARTICLE



Relationships between dynamic risk factors for externalising problem behaviour and group climate in adults with mild intellectual disability in forensic treatment

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Abstract

Background: Relationships between dynamic risk factors for externalising problem behaviour and group climate were investigated in 151 adult in-patients with mild intellectual disability or borderline intellectual functioning in a Dutch secure residential facility.

Method: Regression analysis was used to predict *total group climate score* and *Support, Growth, Repression, and Atmosphere* subscales of the 'Group Climate Inventory'. Predictor variables were *Coping Skills, Attitude towards current treatment, Hostility, and Criminogenic attitudes* subscales of the 'Dynamic Risk Outcome Scales'.

Results: Less hostility predicted a better overall group climate, better support and atmosphere, and less repression. A positive attitude towards current treatment predicted better growth.

Conclusion: Results indicate relationships of hostility and attitude towards current treatment with group climate. A focus on both dynamic risk factors and group climate may provide a basis for improving treatment for this target group.

KEYWORDS

borderline intellectual functioning, dynamic risk factors, forensic treatment, group climate, mild intellectual disability, secure facility

1 | INTRODUCTION

Incidents involving aggression among clients and between clients and staff members is a common and serious problem in institutional settings (Evers et al., 2014; Kelly et al., 2015; Toch & Kupers, 2007). This applies especially to forensic care (Nijman et al., 2006; Omérov et al., 2002). Effective prevention and management of aggression is a top priority in forensic (mental health) settings because inpatient aggression has serious adverse effects on staff and clients, and on society (Kaltiala-Heino et al., 2007; Nijman et al., 2005; Patel & Taylor, 2011).

Several theoretical models have been proposed to explain the development of aggressive behaviour (Daffern et al., 2004; Nijman et al., 1999; Steinert & Whittington, 2013). These models underscore the complex interaction between dispositional characteristics of the client and environmental factors, also referred to as transactional processes (Sameroff, 2009), which contribute to the development of aggressive behaviour (Raine, 2014). Dispositional characteristics involve an individual's emotions, beliefs and values that determine behaviour. Environmental factors are influences from the physical environment and the people around the individual. For example, several studies have found negative associations between the quality of the group climate

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and aggression within institutional settings (De Decker et al., 2018; Neimeijer et al., 2021; Robinson et al., 2018). The group climate refers to the perceived quality of the physical conditions of a ward and the social relationships between its members (Van der Helm, 2019).

Although a relatively large number of studies have been conducted into the relation between the group climate and aggression, there is scant research on the relation between the group climate and dynamic risk factors for externalising problem behaviours (Puzzo et al., 2019). Addressing dynamic risk factors for externalising problems is important as these have also been shown to be predictive of aggression among different groups of clients in both inpatient forensic and non-forensic services (Greer et al., 2020). The present study focuses on the possible relationships between dynamic risk factors for externalising problem behaviour and the perceived group climate in people with mild intellectual disability or borderline intellectual functioning (MID-BIF: IQ 50–85). Although both group climate quality and dynamic risk factors appear to affect the occurrence of inpatient aggression in clients with MID-BIF (see e.g. Drieschner et al., 2013; Neimeijer et al., 2021), there is a gap in our understanding of the relations between dynamic risk factors for externalising problem behaviour and group climate quality in this target group.

Externalising behaviour is an umbrella term for disruptive behaviour that includes multiple outwardly directed behaviours, such as aggression, delinquency, and defiance. For individuals with MID-BIF, externalising problems emerge as one of the most common forms of behavioural problems in forensic facilities (e.g. Delforterie et al., 2018). Static and dynamic risk factors are involved in the development of externalising problem behaviours. Static risk factors such as age are immutable. Dynamic risk factors may be influenced by intervention. Based on the Risk-Need-Responsivity model (Bonta & Andrews, 2017), dynamic risk factors in clients are assessed and targeted in inpatient forensic treatment to tailor treatment to the needs of clients.

Several studies have examined the relations between dynamic risk factors and group climate in youth and adult clients without MID-BIF in secure residential facilities. A study by Van der Helm et al. (2014) found that an open group climate was associated with more adaptive coping skills of clients. Conversely, the authors found that a closed group climate was associated with clients' maladaptive coping skills. An open group climate is characterised by a positive group atmosphere, in which staff members provide growth opportunities by responding to and supporting the needs of their clients. A closed group climate is characterised by a negative group atmosphere, in which staff members generally neglect or manage clients through repressive measures. According to Van der Helm et al. (2012), a closed group climate is also reflected by hostility among clients and staff. Next, a study by Stasch et al. (2018) found that the prison climate perceived by the inmates positively correlated with their attitudes towards treatment. Moreover, a study by Woessner and Schwedler (2014) found that more favourable ratings of prison climate significantly correlated with favourable changes in pro-criminal attitudes, encompassing crime-supportive attitudes and negative attitudes towards the law. Coping skills, treatment attitude, hostility, and criminogenic attitudes are considered dynamic risk factors for

externalising problem behaviour in clients in forensic services (Bonta & Andrews, 2017; Douglas & Skeem, 2005), including clients with MID-BIF (Lindsay et al., 2008; Van der Put et al., 2013).

There is thus some evidence pointing to associations between dynamic risk factors for externalising problem behaviour and group climate in clients without MID-BIF. There are also likely to be similar associations between dynamic risk factors and group climate in clients with MID-BIF (Delforterie et al., 2019). To our knowledge, no research has yet been conducted in this target group into the relationships between dynamic risk factors and group climate. Therefore, primary aim of the current study is to explore relationships between dynamic risk factors for externalising problem behaviours and the quality of the group climate in adults with MID-BIF in a secure residential facility (see the methods section for the hypotheses). Identifying predictors of group climate in this target group could inform interventions aimed at improving the group climate. It can help us better understand and rearrange the group climate in the best way to reduce dynamic risk factors for externalising problem behaviour.

2 | METHODS

2.1 | Setting

We explore the relations between dynamic risk factors for externalising problem behaviour and group climate in adult inpatients with MID-BIF from Trajectum, a forensic residential facility for people with MID-BIF in the eastern and northern parts of the Netherlands. Clients admitted to this facility have severe behavioural disorders resulting from an interaction between their low level of adaptive and intellectual functioning and psychiatric, personality, social and addiction problems. Trajectum holds 33 inpatient wards each with six to 10 beds. Around 75% of these wards have mixed groups of male and female clients, with the remaining wards housing only male clients. These are either high, medium, or low security wards for long-term rehabilitation, and the intensity of care provided to clients also varies from high, medium to low care intensity in these wards. Clients progress from highest to lowest security wards during their treatment. About 85% of the clients in inpatient treatment are men and 15% are women, of whom about 20% are voluntarily admitted, and 80% were admitted by order of the Dutch court for committing criminal offences (Delforterie et al., 2018). The Dutch court can impose forensic treatment as part of a sentence on convicts with a mental disorder or an intellectual disability who are deemed to have a high risk of recidivism, with the aim of working towards their safe reintegration into society without reoffending. In the Netherlands, people can also be voluntarily admitted to a forensic residential facility for psychiatric treatment, so that they are attended to in the event of mental crises that could harm them and others.

Throughout the entire study period, no interventions aimed at specific dynamic risk factors in clients or improvement of group climate were implemented across the 33 wards that participated in the current study. Depending on their needs, clients received courses and therapies in the areas of sexuality, aggression, addiction, and trauma.

2.2 | Participants

Participants were 151 clients from the inpatient treatment units of Trajectum, most of whom were men ($N = 119$; 79%). Their mean age was 37.4 ($SD = 12.30$). IQ scores determined by intelligence tests (e.g., WAIS-IV) were retrieved from client records. Mean total IQ score was 70.3 ($SD = 9.09$), mean verbal IQ 72.3 ($SD = 9.89$) and mean performance IQ 74.5 ($SD = 12.32$). Length of stay was between 2 and 1636 weeks ($M = 177.6$ weeks, $SD = 178.17$). Of the participants, 19% resided in the facility for up to 1 year, 35% from 1 to 3 years, 26% from 3 to 5 years, 19% from 5 to 12 years and 1% for more than 30 years. This wide range in length of stay is inherent to the organisation's unique client base. Clients with MID-BIF often entail longer duration of treatment than clients of average intelligence. In addition to a MID-BIF, clients of Trajectum have various comorbid psychiatric problems and/or distinct forensic backgrounds, resulting in differences between them in length of stay. Clients with different lengths of stay reside together in the rehabilitation wards.

2.3 | Procedure

Data used in this study were collected between December 2015 and April 2019. Group climate data were obtained using the Group Climate Inventory (GCI; Van der Helm et al., 2011) adapted for people with MID-BIF. In this version of the measure, items were edited to match the language skills of the target group, and each answering category was supported visually by a smiley icon (see Neimeijer et al., 2019). Clients were assisted during the administration of the GCI by trained research assistants who read the items and answering categories aloud and, if necessary, explained the items to the clients. The clients gave oral ratings to the items, which were recorded by these research assistants. The research assistants were not involved in the treatment of clients, nor were they stationed in any treatment unit.

Data on dynamic risk factors were obtained using the Dynamic Risk Outcome Scales (DROS; Drieschner & Hesper, 2008), which was administered to therapists via the organisation's Routine Outcome Monitoring system.

Staff members who are working with clients in treatment units on a daily basis and thus help shape the group climate (Sameroff, 2009), were not involved in the data collection process.

A secure Trajectum database containing GCI and DROS data was accessed, from which data were drawn from clients who had both a GCI and DROS measurement performed within a maximum of 3 months interval.

Prior to routine collection of DROS and GCI data, clients were asked to consent to their data being used to provide performance-related feedback to units, as well as for any other research purpose. Conversations were held with clients by their therapists to ensure that clients understood the content of the requested informed consent. Oral informed consent was obtained from all clients. Written informed consent was given by clients, or legal representatives of clients whom the court deemed incapable of granting written consent

TABLE 1 Descriptions of the subscales of the Dynamic Risk Outcome Scales and Group Climate Inventory.

Subscale	Number of items	Content
Dynamic Risk Outcome Scales		
Coping skills	3	Includes coping with conflict, coping with urges to engage in severe problem behaviours, such as revenge and sexual transgressions, and coping with situations one must endure, such as disappointments and loss
Attitude towards current treatment	2	Assesses a client's level of recognition and attitude towards the current need for treatment
Hostility	3	Evaluates the propensity to falsely ascribe hostile intentions to others and respond hostilely, whether there is a latent sense of injustice, and how strong feelings of revenge are and whether they lead to vindictive behaviour
Criminogenic attitudes	4	Assesses attitudes towards physical violence, self-centeredness, antisocial behaviour, and social conventions
Group Climate Inventory		
Support	11	Refers to the support and responsiveness that staff members provide to clients (e.g., "The staff members listen to me when I want to tell something")
Growth	6	Addresses the meaning that clients give to their stay and learning conditions in relation to life inside and outside the facility (e.g., "I get to make my own decisions here")
Repression	7	Measures the amount, fairness, and enforcement of rules, as well as privacy and boredom among clients (e.g., "The staff members always get their way")
Atmosphere	5	Is related to the structure, safety, and trust that clients experience in both the physical and social environment (e.g., "I feel good at the ward")

themselves. Study approval was provided by the Ethics Committee of the Faculty of Social Sciences of the Radboud University (ECSW-2021-084).

2.4 | Instruments

2.4.1 | Group climate inventory

The Group Climate Inventory (GCI) is a self-report measure that can be used to routinely assess the perceived quality of the group climate

	Minimum	Maximum	Mean	SD	Cronbach's α
Dynamic Risk Outcome Scales					
Coping skills	1.00	5.00	2.58	0.78	.83
Attitude towards current treatment	1.00	5.00	3.29	0.94	.88
Hostility	1.00	5.00	3.39	0.87	.78
Criminogenic attitudes	1.00	5.00	3.32	0.78	.84
Group Climate Inventory					
Total group climate score	1.68	4.69	3.42	0.62	.91
Support	1.56	5.00	3.57	0.72	.88
Growth	1.00	5.00	3.58	0.83	.81
Repression	1.29	4.71	2.89	0.70	.69
Atmosphere	1.00	5.00	3.33	0.78	.74

Note: On all variables, a higher score indicates a more favourable outcome, except for Repression, on which a lower score indicates a more favourable outcome.

TABLE 2 Descriptive statistics of the subscales of the Dynamic Risk Outcome Scales and Group Climate Inventory ($N = 151$).

using 29 five-point Likert-type items (1 = “strongly disagree” to 5 = “strongly agree”) covering the four subscales of Support, Growth, Repression and Atmosphere (Van der Helm et al., 2009, 2011). The GCI has demonstrated good psychometric properties in several client populations from institutional and correctional facilities (Heynen et al., 2014; Strijbosch et al., 2014, 2018; Van der Helm et al., 2011), including adults with MID-BIF (Neimeijer et al., 2019).

All subscales generate a separate mean score between 1 and 5, on which a higher score is more favourable, except for the Repression subscale, on which a lower score is more favourable. All GCI items combined (after Repression items were reverse coded) yield a total group climate scale score between 1 and 5, of which a higher score indicates a more open group climate and a lower score a more closed group climate. The descriptions of the GCI subscales are shown in Table 1. Internal consistencies of the GCI subscales and total scale are shown in Table 2.

2.4.2 | Dynamic risk outcome scales

The Dynamic Risk Outcome Scales (DROS) is a therapist-reported measure developed specifically for people with MID-BIF that maps dynamic risk factors for externalising problem behaviour (Drieschner & Hesper, 2008). It comprises 42 items across 14 subscales and is completed every 6 months by a client's therapist. DROS items have answering categories on a five-point scale. Each item has a specific description (i.e., anchor) of client cognitions or behaviour at the answering options 1, 3 and 5. Options 2 and 4 are in-between answers when the specific descriptions at 1, 3 or 5 are not fully applicable to the client. A score of 5 indicates the absence of risk, while a score of 1 indicates the highest risk of externalising behaviour problems for each DROS item. Each subscale produces an average score between 1 and 5, with a higher score being more favourable. The DROS has shown an acceptable factorial structure and good internal consistency for most subscales (Delforterie et al., 2018).

In the present study, we included data from four DROS subscales covering a total of 12 items, namely ‘Coping skills’, ‘Attitude towards current treatment’, ‘Hostility’ and ‘Criminogenic attitudes’. These DROS subscales were included as they may be related to group climate based on the studies (as mentioned in the introduction) that reported associations between group climate and coping skills (Van der Helm et al., 2014), hostility (Van der Helm et al., 2012), attitude towards treatment (Stasch et al., 2018), and pro-criminal attitudes (Woessner & Schwedler, 2014). The descriptions of the DROS subscales are shown in Table 1 and the internal consistencies in Table 2.

2.5 | Hypotheses

We hypothesise a positive relationship between the total group climate score and the DROS subscales of ‘Coping skills’, and ‘Attitude towards current treatment’. In addition, we presume an inverse relationship between the total group climate score and the subscales of ‘Hostility’ and ‘Criminogenic attitudes’. Moreover, we hypothesise positive relationships of group climate subscales of *support*, *growth* and *atmosphere* with Coping skills and Attitude towards current treatment, and of *repression* with Hostility and Criminogenic attitudes. Furthermore, we hypothesise negative relationships of support, growth and atmosphere with Hostility and Criminogenic attitudes, and of repression with Coping skills and Attitude towards current treatment.

2.6 | Statistical analyses

A hierarchical linear regression analysis (Landau & Everitt, 2004) was used to examine the relationships of the GCI total group climate score and subscale scores with each of the four subscales from the DROS as predictor variables. We controlled for gender, age and treatment duration as these variables were reported to influence change in dynamic risk factors in clients from admission to release in a study by Van der Linde et al. (2020). All statistical procedures

TABLE 3 Pearson correlation coefficients between the subscales of the Dynamic Risk Outcome Scales and Group Climate Inventory (N = 151).

	Coping skills	Attitude towards current treatment	Hostility	Criminogenic attitudes	Total group climate score	Support	Growth	Repression	Atmosphere
Coping Skills	–								
Attitude towards current treatment	.510*	–							
Hostility	.482*	.573*	–						
Criminogenic attitudes	.507*	.603*	.630*	–					
Total group climate score	.454*	.413*	.492*	.358*	–				
Support	.433*	.394*	.481*	.369*	.932*	–			
Growth	.329*	.435*	.382*	.365*	.820*	.735*	–		
Repression	–.408*	–.384*	–.420*	–.260*	–.815*	–.643*	–.581*	–	
Atmosphere	.277*	.065	.277*	.108	.637*	.506*	.269*	–.452*	–

Note: On all variables, a higher score indicates a more favourable outcome, except for Repression, on which a lower score indicates a more favourable outcome. * $p < .001$ (2-tailed).

were performed in SPSS Version 28.0. Mean time between DROS and GCI administration was 1.37 months (SD = 1.00). Missing values were 0.56% (N = 37 out of 6644 possible values) across 44 variables (i.e., 29 GCI items, 12 DROS items, gender, age, and length of stay). There was a random pattern to the missing values (Little's MCAR test: $p = 0.674$), meaning that these data points were missing, independent of the observed and missing data. According to Dong and Peng (2013), potential biases caused by missing data of this magnitude on the results of the current study would be negligible.

Data obtained with GCI and DROS are used to monitor performance indicators for achieving certain goals over time aimed at improving forensic treatment (Delforterie & Hesper, 2020; Van der Helm, 2019). As the GCI is deployed in the field, the total group climate score and GCI subscale scores are only computed for clients who have completed at least 80% of all items and that of the items within subscales, respectively (Roest et al., 2016). A DROS subscale score is computed by averaging the number of items completed in that subscale. If only one item is completed within a subscale, the value of that item is used when providing performance-related feedback to the therapist on that subscale.

To minimise potential biases due to missing data, we only included clients who completed at least 80% of all GCI items and 80% of the items within each GCI subscale, and for whom at least 75% of the items within each DROS subscale were completed. Eight of 159 clients for whom data had been drawn were therefore excluded, resulting in a sample of 151 clients.

As we test with eight predictors (including the Intercept 'constant', gender, age, length of stay, and four DROS subscales) in each two-block regression model, we apply Bonferroni correction as

recommended by Mundfrom et al. (2006) to counteract the potential bias due to multiple testing. With eight tests being considered, we have $\approx 34\%$ chance ($1 - (1 - \alpha)^k$) of finding at least one Type I error. To eliminate this probability, the p value must be less than .006 (α/k) to reject a null hypothesis. The significance level is therefore set at $\alpha = 0.006$.

3 | RESULTS

Table 2 summarises the minimum, maximum and mean values, standard deviations and internal consistencies of the DROS and GCI subscales.

3.1 | Analyses

Using Pearson's r , we calculated linear correlations between the DROS and the GCI scores (Table 3). As we found significant small to medium-sized correlations between virtually all GCI and DROS variables, we examined the relations between the DROS subscale scores and the total group climate score by means of a hierarchical linear regression analysis, controlling for gender, age, and length of stay.

Data were first tested against the assumptions for a hierarchical linear regression analysis. An analysis of standard residuals showed that the data contained no outliers (Std. Residual Min = -2.45 , Std. Residual Max = 2.36). In addition, the data met the assumption of collinearity (Coping skills, Tolerance = 0.61, VIF = 1.64; Attitude towards current treatment, Tolerance = 0.52, VIF = 1.92; Hostility, Tolerance = 0.53, VIF = 1.90; Criminogenic attitudes, Tolerance = 0.46,

VIF = 2.16) and that of independent errors (Durbin-Watson = 1.91). Judging from the histogram and the normal P-P plot of standardised residuals, the data followed an approximately normal distribution. We also tested whether the assumptions of homogeneity of variance and linearity were met by assessing the scatterplot of standardised predicted values. Furthermore, the data met the assumption of non-zero variances (Gender, Variance = 0.17; Age, Variance = 151.23; Length of stay, Variance = 31745.69; Coping skills, Variance = 0.61; Attitude towards current treatment, Variance = 0.89; Hostility, Variance = 0.76; Criminogenic attitudes, Variance = 0.61; Total group climate score, Variance = 0.39). As the data met all required assumptions, a hierarchical linear regression analysis was warranted to evaluate the prediction of the total group climate score from the DROS subscales Coping skills, Attitude towards current treatment, Hostility, and Criminogenic attitudes.

Gender, age, and length of stay were included as control variables in the first block analysis, which revealed a statistically significant model ($p = .026$). Moreover, the R^2 value of 0.062 associated with this regression model indicates that gender, age, and length of stay accounted for 6.2% of the variation in the total group climate score. The proportion of variance explained by the control variables is therefore limited.

Coping skills, Attitude towards current treatment, Hostility, and Criminogenic attitudes were added as variables in the second block analysis. This revealed a model that was statistically significant, meaning that gender, age, length of stay and the dynamic risk factor variables together accounted for a statistically significant variation in the total group climate score. Using the enter method, we found that these variables explained 35% of the variation in the total group climate score ($F(7, 141) = 10.69, p < .001, R^2 = 0.35$, Adjusted

$R^2 = 0.31$; R^2 change = 0.29), which means that the addition of the DROS variables contributes to 29% additional variance in the total group climate score. A moderate proportion of the variance in the total group climate score is thus explained by the DROS variables. The Cohen's $f^2 = 0.54 (R^2/1 - R^2)$, indicating a large effect of all modelled predictors on the total group climate score, as effect sizes are considered small if $f^2 \geq 0.02$, medium if $f^2 \geq 0.15$ and large if $f^2 \geq 0.35$ (Cohen, 1988; Selya et al., 2012).

The analyses showed that Hostility significantly predicted the total group climate score. No significant effects were found for the other DROS variables (Table 4).

3.1.1 | GCI subscale analyses

We performed analyses to examine which GCI subscale scores were significantly predicted by the DROS variables. The data met the assumptions for hierarchical linear regression analysis with each GCI subscale as dependent variable (results of the assumption tests are available from the first author). Gender, age, and length of stay were included as control variables in each first block analysis. Coping skills, Attitude towards current treatment, Hostility, and Criminogenic attitudes were added in every second block analysis, resulting in a statistically significant model for each GCI subscale. Using the enter method, the DROS variables explained 13% to 27% additional variation in the values of the GCI subscales (Table 5).

The analyses revealed that Attitude towards current treatment significantly predicted the value of Growth. Furthermore, Hostility significantly predicted the values of Support, Repression and Atmosphere (Table 6).

TABLE 4 Regression coefficients, t values, degrees of freedom and p values from the second block analysis of the subscales of the Dynamic Risk Outcome Scales as predictor variables for the total group climate score ($N = 151$).

Predictor variable	Unstandardised Coefficients		Standardised Coefficients	t value	Degrees of freedom	p value
	β	Standard error	β			
Coping skills	0.17	0.07	0.21	143	2.36	.019
Attitude towards current treatment	0.11	0.06	0.18	143	1.85	.067
Hostility	0.24	0.07	0.34	143	3.62	< .001*
Criminogenic attitudes	-0.07	0.08	-0.09	143	-0.87	.388

Note: * $p < .006$.

TABLE 5 Model summary of the second block analysis for each subscale of the Group Climate Inventory.

	F value	Degrees of freedom	R^2	Adjusted R^2	R^2 change	p value	Cohen's f^2
Support	9.85	7. 141	.33	.30	.27	<.001	.49
Growth	7.08	7. 141	.26	.22	.20	<.001	.35
Repression	7.39	7. 141	.27	.23	.23	<.001	.37
Atmosphere	4.20	7. 141	.17	.13	.13	<.001	.20

Note: First block predictors: (constant), gender, age, and length of stay in weeks; Second block predictors: (constant), gender, age, length of stay in weeks, coping skills, attitude towards current treatment, hostility, and criminogenic attitudes; Model summary of the first block analyses is available from the first author.

TABLE 6 Regression coefficients, *t* values, degrees of freedom and *p* values from the second block analyses of the subscales of the Dynamic Risk Outcome Scales as predictor variables for the Support, Growth, Repression and Atmosphere subscales of the Group Climate Inventory (*N* = 151).

	Unstandardised coefficients		Standardised coefficients	t value	Degrees of freedom	p value
Predictor variable	β	SE	β			
Support						
Coping skills	0.19	0.08	0.20	2.22	143	.028
Attitude towards current treatment	0.11	0.07	0.15	1.51	143	.133
Hostility	0.27	0.08	0.33	3.45	143	< .001*
Criminogenic attitudes	−0.04	0.10	−0.04	−0.38	143	.702
Growth						
Coping skills	0.06	0.10	0.05	0.58	143	.563
Attitude towards current treatment	0.27	0.09	0.31	3.11	143	.002*
Hostility	0.12	0.10	0.13	1.30	143	.197
Criminogenic attitudes	0.03	0.12	0.03	0.27	143	.790
Repression						
Coping skills	−0.18	0.08	−0.20	−2.21	143	.029
Attitude towards current treatment	−0.16	0.07	−0.22	−2.20	143	.029
Hostility	−0.24	0.08	−0.30	−3.01	143	.003*
Criminogenic attitudes	0.17	0.10	0.19	1.76	143	.080
Atmosphere						
Coping skills	0.23	0.10	0.23	2.29	143	.023
Attitude towards current treatment	−0.14	0.09	−0.16	−1.54	143	.127
Hostility	0.32	0.10	0.35	3.35	143	.001*
Criminogenic attitudes	−0.13	0.11	−0.12	−1.10	143	.273

Note: *p* < .006.

4 | DISCUSSION

This study investigated relationships between four dynamic risk factors for externalising problem behaviour and group climate in adults with MID-BIF from inpatient forensic treatment. As the DROS was a proxy measure whose items were rated by therapists, and the GCI items were self-rated by clients, both measures were independent of each other. We hypothesised a positive relationship of overall group climate quality and of *support*, *growth*, and *atmosphere* with ‘Coping skills’ and ‘Attitude towards current treatment’, and of *repression* with ‘Hostility’ and ‘Criminogenic attitudes’. We also hypothesised a negative relationship of overall group climate quality and of support, growth and atmosphere with Hostility and Criminogenic attitudes, and of repression with Coping skills and Attitude towards current treatment. All significant relationships found were in the hypothesised direction. The findings suggest that Hostility is significantly related to the overall quality of the group climate and Support, Repression and Atmosphere in this sample of MID-BIF clients, but not to Growth. In addition, Attitude towards current treatment was significantly related to Growth, but not to the overall quality of the group climate and other subscales.

4.1 | Implications of the results

Although no studies have directly examined the associations between hostility and group climate, our finding that Hostility is significantly negatively related to group climate quality is consistent with reports that hostility is common in wards with a closed group climate (Van der Helm et al., 2012). Clients residing in those wards may show hostility because they are more exposed to unfair treatment, prejudice, victimisation, and harassment for extended periods of time (Austin & Irwin, 2001; Verstegen et al., 2022). In addition, staff members are quicker to respond with repressive measures against clients who exhibit hostile behaviour, thereby further increasing hostility among these clients (De Valk, 2019).

In line with previous research (Stasch et al., 2018; Woessner & Schwedler, 2014), we found small to medium-sized correlations of dynamic risk factors Attitude towards current treatment and Criminogenic attitudes with group climate variables. Our study further indicates that Attitude towards current treatment is significantly related to Growth. A possible explanation for this significant relation is that clients with a more positive attitude towards the treatment are more likely than clients with a more negative attitude to continue with their

treatment and therefore achieve better treatment outcomes (Hagger et al., 2017). In addition, a more positive client attitude could pave the way for a deeper client-caregivers relationship (Tryon et al., 2018), in which a client is more likely to experience growth opportunities.

In contrast to the findings of the study by Van der Helm et al. (2014), we found no significant relationships between Coping skills and group climate variables. The lack of effects in our study could be due to several reasons. Besides using a different measure of coping skills, our sample and context differ as they focused on 59 juvenile delinquents from a youth correctional facility. In addition, different methods used are another likely reason, as study results have been shown to be affected even when different statistical analyses are performed on the same dataset (Silberzahn et al., 2018).

4.1.1 | Limitations

There were two notable limitations to this study. First, due to the limitation of a retrospective cross-sectional study design, we cannot infer causality beyond the associations found between dynamic risk factors and group climate outcomes. Second, as data were collected over a three-and-a-half-year period in which the DROS and GCI were not administered concurrently, but at a maximum interval of 3 months, this may raise concerns about the validity of our findings. It may be that changes have occurred in the wards within 3 months or that clients have switched wards, which may have resulted in an overestimation of the relationships found in this study. In spite of these obvious limitations, this study adds to the scarce body of literature on group climate research in people with MID-BIF.

4.2 | Applications of the results to policy and practice

The results suggest that Hostility and Attitude towards current treatment are related to group climate. As these indicate bidirectional relationships (Sameroff, 2009), a focus on both dynamic risk factors and group climate may provide a basis for improving the treatment for adult forensic inpatients with MID-BIF. Research that has shown beneficial effects of interventions on these dynamic risk factors and on the group climate may be worth mentioning, as it could provide direction to future interventions for this target group.

4.2.1 | Interventions targeting hostility and attitude towards current treatment

There have been reported beneficial effects of dialectical behaviour therapy (Swales, 2020) on hostility in clients without MID-BIF. A study by Evershed et al. (2003) found that male adult forensic clients who participated in a dialectical behaviour therapy aimed at reducing anger and violence showed significantly better outcomes on hostility

and anger than controls. A study by Long et al. (2010) showed that adult women from secure wards who completed 12 sessions of a dialectical behaviour therapy targeting distress tolerance, emotional regulation, and to reduce emotional vulnerability, scored significantly more favourably on client hostility at post-treatment follow-up compared with those who completed less than 12 sessions. Dialectical behaviour therapy may also benefit clients with MID-BIF (McNair et al., 2016).

Compliance therapy (Kemp et al., 1996) can positively influence clients' attitudes towards treatment because it focuses on explaining the benefits of treatment and resolving any ambivalence clients may have about treatment (Gray et al., 2016). It is based on motivational interviewing and cognitive-behavioural therapy (Barrera, 2017; Frielink & Embregts, 2013). Favourable outcomes of compliance therapy have been reported in clients with psychotic disorders on attitudes towards medication (Cavezza et al., 2013; Maneesakorn et al., 2007) and attitudes towards the need for treatment (Hayashi et al., 2001).

4.2.2 | Group climate interventions

Creating and maintaining a therapeutic and nurturing group climate requires that staff members foster supportive relationships with clients, provide clients with plenty of growth opportunities, and maintain a low level of repression. Only by meeting these requirements can treatment in forensic facilities be provided in a safe atmosphere in which the psychological needs of clients can be met (Van der Helm, 2019). To establish a favourable group climate in which learning is optimal for clients with MID-BIF, it is recommended that forensic facilities invest in continuous staff training.

Providing psychoeducation could make staff members more aware of their roles in supporting a positive group climate, which in turn may result in clients assessing the perceived group climate more favourably. For example, a study by Nessel et al. (2009) reported significant improvements in the perceptions of group climate of adult clients in a forensic psychiatric clinic after staff members completed a three-week psychoeducation programme that focused on the group climate.

In a review by Dickens et al. (2022), interventions aimed at improving relationships between staff and clients showed the most promise in specifically improving group climate in acute mental health inpatient settings. Skills training to improve interactions between staff and clients may also benefit the group climate perceptions of clients with MID-BIF. For example, a study by Visser et al. (2021) showed significant beneficial changes in the group climate perceptions of youth clients with MID after staff members were trained to utilise Non-Violent Resistance (Omer & Lebowitz, 2016) adapted for MID (NVR-MID). The NVR-MID aims to correct problematic behaviour in clients with MID through a variety of non-escalatory methods used by trained staff members, such as announcements in which friendly written statements are provided to clients that certain behaviours are not acceptable (for the other NVR-MID methods, see Visser et al., 2021).

It may also be beneficial to the group climate to educate staff about the difficulties of working with clients with MID-BIF in forensic settings and train them to effectively manage these difficulties. A study by Doyle (2017) in a low security inpatient forensic mental health unit reported beneficial effects of a staff-centred adaptation of mentalization based treatment on the maintenance of a positively perceived group climate by clients and staff and on the resilience of staff (i.e. the stress levels experienced by the staff and their attitude towards working with clients with personality disorders). This staff-centred intervention consisted of skills and awareness training, combined with group sessions led by an external coach focused on working with clients who present with difficulties in interpersonal relationships.

Repression by staff is related to aggressive behaviour in clients with MID-BIF and represents a major impediment to successful forensic residential treatment (De Valk, 2019; Neimeijer et al., 2021). Therefore, it is necessary to reduce coercive measures such as physical restraint, seclusion and chemical restraint and use less restrictive means to minimise aggressive behaviour and maximise treatment efficacy. To introduce practice initiatives involving less coercive measures, staff members need to be educated about the causes of aggressive behaviour in clients and understand factors that may preempt or exacerbate aggressive behaviour (Ewington, 2016). An intervention with the potential to reduce repression in inpatient treatment, and conflicts between staff and clients, is the Safewards Model (see Dickens et al., 2022). The Safewards Model is an adaptable programme consisting of 10 interrelated evidence-based interventions that target both staff and clients. One of these is called mutual help meetings, which refers to the organisation of frequent meetings on the ward in which staff and clients are encouraged to identify ways to support each other (for the other Safewards interventions, see e.g. Baumgardt et al., 2019).

The best results in terms of reducing repression could probably be achieved if additional efforts were made to improve the functioning of staff in treatment units. A study by Neimeijer et al. (2022) found a relationship between a well-functioning team as perceived by staff (i.e., there is good communication and collaboration, a clear division of tasks, and mutual trust and support in a team) and less perceived repression by clients with MID-BIF in a forensic setting. Improving the functioning of teams may have more beneficial effects than on clients' perceptions of repression alone, as the authors also found that perceived workload and poor team functioning by staff were related to fewer perceived growth opportunities by clients. Based on Neimeijer et al. (2022), investing in team coaching for forensic settings is recommended. Team coaches could help teams to spread the workload more evenly among staff, but also to reduce the 'integrity paradox' and subsequent dehumanisation of clients (De Valk, 2019). The integrity paradox is proposed by Ellemers and De Gilder (2022) and indicates that highly motivated staff confronted by aggression or other negative behaviour, develop countertransference and sometimes repressive behaviour because they perceive themselves as being prevented from 'doing the good work'. In addition, training the leadership skills of team managers could be beneficial for the functioning of teams (Van der Helm et al., 2018).

5 | CONCLUSION

This study shows how the outcomes of dynamic risk factors in clients may impact organisational issues in forensic inpatient units, and vice versa, and in particular demonstrates bidirectional relationships of hostility and attitude towards current treatment with group climate. Policy needs to focus on opportunities to increase awareness, knowledge and skills of staff surrounding the dynamic risk factors of clients and the group climate, as well as to improve the functioning of teams of staff in units to achieve optimal treatment for MID-BIF clients. However, due to methodological shortcomings of this study, our results should be interpreted with caution. Future research is needed on the relationships between dynamic risk factors and group climate in inpatient forensic treatment to determine the direction of the relationships between the variables and to evaluate the validity of the results. Future studies should aim to improve upon the current study by adopting a prospective longitudinal study design with different measurement points in time to establish causality between dynamic risk factors and group climate outcomes. In addition, shortening the time between administrations of the DROS and GCI under more controlled conditions could reduce potential bias due to the elapsed time between the two measurements in future studies.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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